Kahawa Roots

Traceability Platform

Tracking Coffee from Kenya's Farms to the UK

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Pre-Interview Task — Developer Intern — April 2026

Proof: I've Built This Before

My live full-stack project on the **same GCP infrastructure** I'm proposing:

- DeltaEd — Live on GCP Cloud Run
- GCP Cloud Run, `europe-west2` (London)
- Dockerized, full stack
- React + Next.js + PostgreSQL

happy to walk through the architecture if interested.

Part 1 & 3: How I'd Build This

How Data is Stored & Recorded

- every shipment gets a unique ID when its first recorded
- raw data (form/CSV) goes straight into **GCP Cloud Storage** as a backup — never touched, never deleted
- a **Cloud Function** auto-triggers, cleans and validates the data
- clean data lands in **Cloud SQL (PostgreSQL)** — 3 tables: `farmers`, `shipments`, `shipment_stages`
- each stage (farm → processing → export → UK) is a row in `shipment_stages` with timestamps and GPS

QR Code → Webpage

Non-Technical Data Entry (Part 3)

the founders (Rupal & Cetric) aren't developers — so the system has two simple ways to add data:

Option 1: Web Form

- fill in a simple form on the admin dashboard
- one shipment at a time
- no coding, no terminal, no SQL

Option 2: CSV/Excel Upload

- drag and drop a spreadsheet

Part 2: Technology Choices + Bonus

Tech Stack & Why

Component	Choice & Why
Cloud	**GCP** (London region) — **up to $200K startup credits** — UK data centre = GDPR friendly — Cloud Run scales to zero = pay nothing when idle
Backend	**Python + REST API** on Cloud Run — fast to build, great for data validation
Frontend	**Next.js (React)** on Cloud Run — SSR for fast QR page loads, SEO friendly
Database	**Cloud SQL (PostgreSQL)** — relational, sub-10ms reads for

Data Architecture: Medallion Pattern

Bronze Layer
Raw CSV/JSON in Cloud Storage

↓

Silver Layer
Clean data in PostgreSQL

↓

Gold Layer
QR Pages + BigQuery (future)

Bonus: Database Schema

```
CREATE TABLE shipments ( id UUID PRIMARY KEY,
batch_id VARCHAR(50) UNIQUE, farmer_id UUID REFERENCES farmers(id), grade VA
AA, AB, PB weight_kg DECIMAL, status VARCHAR(20), created_at TIMESTAMP DEF
CREATE TABLE
shipment_stages(id UUID PRIMARY KEY, shipment_id UUID REFERENCES shipments
farm, processing, — —
export, import location VARCHAR(100), timestamp TIMESTAMP, notes TEXT);
```

System Architecture

YOUR ARCHITECTURE DIAGRAM HERE

Export from draw.io as PNG
and replace this placeholder

File → Export as → PNG (2x scale)
Save as `architecture.png`

Click here for interactive architecture diagram